

**Guidelines for Discipline Specific Core Semester VII
(NEP-UGCF 2022)
Compiler Design
(Effective for Academic Year 2025-26)**

Sr. No.	Topics	Chapter	No. of Hours	Reference
1	UNIT 1: Overview of compiler, Phases of compiler	1.1, 1.2	2	[1]
2	UNIT 2: Lexical analysis: role of Lexical analyser, specifications and recognition of tokens, symbol table, error reporting, regular expressions and definitions, lexical analyser - LEX	3.1, 3.2, 3.3, 2.7, 3.4(3.4.1, 3.4.2, 3.4.3), 3.5,	9	[1]
3	UNIT 3: CFGs, left recursion, left factoring, Top down parsing - LL parser, Bottom-up parsing - LR parser, Parser generator -YACC	4.1.1, 4.1.2, 4.1.3, 4.2(4.2.1 to 4.2.5), 4.3, 4.4, 4.5, 4.6, 4.9	15	[1]
4	UNIT 4: Syntax directed definitions, evaluation orders for syntax directed definitions, Intermediate languages: syntax tree, three address code, Types and declarations, translation of expressions, loops and conditional statements, type checking	5.1, 5.2 6.1.1, 6.2(6.2.1 to 6.2.3), 6.3(6.3.1 to 6.3.5), 6.4, 6.5(6.5.1 & 6.5.2), 6.6(6.6.1 to 6.6.4)	10	[1]
5	UNIT 5: Activation records, stack allocation, Issues in code generation - design of a code generator	7.1, 7.2 8.1, 8.2, 8.3(only introduction), 8.4 (8.4.1 to 8.4.3)	6	[1]
6	UNIT 6: Principal sources of optimization, peephole optimization	8.7	3	[1]

References

Text Books

1. Aho, A., Lam, M., Sethi, R., & Ullman, J. D. (2006). *Compilers: Principles, Techniques, and Tools*. 2nd edition. Addison Wesley.

<https://theswissbay.ch/pdf/Gentoomen%20Library/Programming/Compiler/Aho%20-%20Compilers%20-%20Principles%2C%20Techniques%2C%20and%20Tools%20e.pdf>

Reference Books

1. V Raghvan (2010), "Principles of Compiler Design", TMH.
2. Santanu Chattopadhyay (2005), "Compiler Design", PHI.

Suggested Practical List:

1. Write a Lex program to count the number of lines and characters in the input file.
2. Lex program to count the number of vowels and consonants in a given string
3. Write a Lex program that implements the Caesar cipher: it replaces every letter with the one three letters after in alphabetical order, wrapping around at Z. e.g. a is replaced by d, b by e, and so on z by c.
4. Write a Lex program that finds the longest word (defined as a contiguous string of upper and lower case letters) in the input.
5. Write a Lex program that distinguishes keywords, integers, floats, identifiers, operators, and comments in any simple programming language.
6. Write a Lex program to count the number of words, characters, blank spaces and lines in a C file.
7. Write a Lex specification program that generates a C program which takes a string "abcd" and prints the following output:

abcd

abc

ab

a

8. A program in Lex to recognize a valid arithmetic expression.
9. Write a YACC program to find the validity of a given expression (for operators + - * and /)A program in YACC which recognizes a valid variable which starts with letter followed by a digit. The letter should be in lowercase only.
10. A Program in YACC to evaluate an expression (simple calculator program for addition and subtraction, multiplication, division).

11. Program in YACC to recognize the string „abbb“, „ab“ „a“ of the language $(a^n b^n, n \geq 1)$.
12. Program in YACC to recognize the language $(a^n b, n \geq 10)$. (output to say input is valid or not)